

Biplov Singh

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PROFESSIONAL SUMMARY

Data Engineer specializing in real-time streaming architectures, distributed processing, and open-source data orchestration. Contributor to Apache Airflow with strong proficiency in PySpark, Apache Kafka, SQL, Docker, and AWS Terraform. Passionate about building robust, high-throughput ETL pipelines and cloud data infrastructure.

EDUCATION

- **Masters in Data Science** 2025-27
Indian Institutes of Information Technology, Lucknow
- **Bachelor of Science in Computer Application and Cyber Security** 2022-25
Jharkhand Raksha Shakti University, Ranchi CGPA: 7.92

OPEN SOURCE CONTRIBUTIONS

- **Apache Airflow** Jun 2026 – Present
Contributor / Python, pytest, Breeze, CI/CD 🔗 Pull Requests
 - Contributed code modifications to the core repository of Apache Airflow, standardizing provider testing infrastructures.
 - Refactored provider test suites for **Snowflake** (#68898) and **Teradata** (#68190) modules, eliminating legacy **unittest** imports in favor of **pytest**.
 - Validated pipeline stability by passing 80+ continuous integration checks across multi-platform test runners and AMD environments.

PROJECTS

- **Real-Time Ride Analytics Pipeline** March 2026 – June 2026
Production-grade streaming architecture for real-time geospatial processing and surge detection.
 - Architected a high-throughput event streaming pipeline using Apache Kafka and PySpark Structured Streaming to process 100+ simulated geospatial events per second.
 - Containerized the comprehensive microservices environment using Docker Compose and automated AWS cloud infrastructure deployment via Terraform.
- **Clickstream Analytics Pipeline** Jan 2026 – Feb 2026
End-to-end big data pipeline for processing website clickstream data using Hadoop.
 - Built a complete ETL pipeline using Apache Flume, Pig, and Hive to process website clickstream data.
 - Designed data ingestion workflows to collect and stream raw user interaction logs into the Hadoop ecosystem.
 - Containerized the pipeline environment using Docker for reproducible deployment and testing.

ACADEMIC EXPERIENCE

- **Teaching Assistant** Aug 2025 - Dec 2025
Indian Institutes of Information Technology, Lucknow Lucknow, UP
 - Assisted professors in managing coursework, designing assignments, and delivering supplementary instruction for undergraduate computer science and data science courses.
 - Conducted weekly doubt-clearing and practical lab sessions, guiding students through complex programming concepts and debugging.
 - Evaluated and graded student projects, assignments, and examinations, ensuring fair and constructive feedback to support academic growth.

TECHNICAL SKILLS AND INTERESTS

Languages: Python, SQL, C/C++, JavaScript

Libraries: Pandas, Numpy, PyTorch, Matplotlib, Scikit-Learn, Git, Docker, Kubernetes, pytest

Data Engineering & Big Data: Hadoop, Apache Hive, Apache Pig, Apache Flume, Apache Kafka, PySpark, YARN, ETL Pipelines

Operating System: Linux/Ubuntu

Cloud/Databases: PostgreSQL, MongoDB, Terraform, AWS, Azure

Areas of Interest: Data Engineering, Big Data Analytics, Distributed Systems, Machine Learning, Cloud Computing